Way to go

Reservation application for traveling

Software Requirements Specification

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1. Introduction

*This section outlines the contents of the SRS document for the **WayToGo travel app**. Also, the purpose for this document is described and a list of abbreviations and definitions is provided.*

1.1 Purpose

The goal of designing the travel services application is to provide easy and convenient solutions for travelers to book their tickets and manage their trips quickly and smoothly. The application aims to offer a simple and flexible user interface that caters to the needs of all users easily, ensuring an enjoyable and seamless travel experience. Additionally, some constraints may impact the functionality or user experience.

1.2 Scope

***WayToGo** is an application designed to facilitate access to travel services for individuals of all abilities. Users can conveniently access services anytime, anywhere, with seamless booking experiences. Through partnerships with airlines, **WayToGo** offers a wide range of services, including flight and hotel reservations, car rentals, and find a tour guide ensuring comprehensive travel solutions. Additionally, it provides users with a curated list of events at their destination. **WayToGo** prioritizes customer satisfaction, striving to deliver exceptional travel booking experiences for all users.*

1.3 Intended Audience and Intended Use

The application is designed to cater to all users who want to travel, book tickets, and access other services in a seamless, fast, and user-friendly manner.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definitions
App	Is a shortened form of "application," which refers to a software program designed to perform a specific task or set of tasks on a computer or mobile device.
User	Someone who interacts with the mobile phone application.
Booking	Refers to the process of reserving and securing a service or accommodation for a specific date or time.
Seamless	Smooth and continuous, without any interruptions or gaps.
Caters	To provide what is needed or desired; to satisfy the needs or wishes of someone.
Comprehensive	Covering or including everything; having all aspects or elements.
Curated	Carefully selected or organized, often with specific criteria in mind.
Prioritizes	Gives importance or preference to; treats as more important or urgent.

Visually impaired	Having partial or complete loss of vision; experiencing difficulty in seeing.
Absher	Absher is a system used in Saudi Arabia that allows users to access various government services, including digital identification and authentication.
Fulfill	To meet the requirements or conditions; to carry out or satisfy an obligation.
Destination	The place to which someone is going or something is being sent.
Tourist attractions	Places or sites that are popular for tourists to visit due to their cultural, historical, or natural significance.
Administrator	A person is responsible for managing and overseeing the operations or affairs of an organization or system.
Dependency	Reliance on something else, often referring to the need for an Internet connection for functionality.
Database	A structured collection of data stored electronically in a computer system.
Protocols	Rules governing how devices communicate online.
Encryption	Converting data into a coded format to prevent unauthorized access, safeguarding sensitive information.
Data Transfer Rates	Speed of data transfer between devices online, measured in bps, kbps, or mbps, impacting upload speed and user experience.
iOS devices	Refers to Apple's mobile devices such as iPhones and iPads.
RAM	Refers to having a large amount of Random Access Memory (RAM), which is essential for running applications smoothly.
Facial recognition	Biometric technology that identifies or verifies individuals based on their facial features, used for secure access control in the application.
NAFATH	An application that allows unified central access to all service providers by verifying the user's identity.
GPS (Global Positioning System)	A satellite-based navigation system that enables precise determination of location and time anywhere on Earth.
Java SE	Specifies the version of Java Standard Edition (SE) used to develop the application's codeز
HTTP (Hypertext Transfer Protocol)	Protocol used for transmitting data over a computer network.
HTTPS (Hypertext Transfer Protocol Secure)	Protocol that adds a layer of security (SSL/TLS encryption) to HTTP, ensuring secure communication over a computer network.
SSL/TLS (Secure Sockets Layer/Transport Layer Security)	Protocols used to provide secure communication over a computer network, ensuring data privacy and integrity.
Error Handling	Dealing with problems that occur, such as helping users recover a lost password or managing duplicate accounts

API (Application Programming Interface)	An API is like a special code that lets different programs talk to each other and share information. It's a way for software to work together without needing to know all the details inside each program.
Accommodation	A place where travelers stay temporarily, like a hotel room, apartment, or villa.
Venue	The desired location where the event will be hosted.
Data Integrity	Making sure information stays accurate and reliable over time, without being changed or corrupted.
Scalability	Building systems or apps that can handle more users or data without slowing down or breaking.
Backup and Recovery	Making copies of important information regularly so it can be restored if something goes wrong like a computer crash.

1.5 References

- *Booking.com. (n.d.). Hotels, Apartments, Villas & More. Retrieved from https://www.booking.com/*
- *Almatar.com. (2024). Book Hotels & Flights at The Best Price. Retrieved from https://almatar.com/ar/*
- *Almosafer. (2024). Flights, Hotels, Activities & Airlines Ticket Booking. Retrieved from https://www.almosafer.com/en/.*

2. Overall Description

WayToGo *is more than just an application, simplifying the entire journey process from booking flights, and taxis, to hotels. With support for multiple languages and specialized services for the visually impaired, it ensures an easy, efficient, and organized experience, perfect for first-time travelers.*

2.1 Product Perspective

This document is crafted from the viewpoint of the users of the ***WayToGo*** *application, who utilize it to streamline and manage their travel arrangements seamlessly.*

2.2 Product Functions

1. ***Sign in:*** *the user may sign in by username and password or by Absher.*
2. ***Renting a car****: determine the location to receive the car, the location of its return, determine the duration of the rental, and request the driver's data to confirm the fulfillment of the conditions for driving by the law of the country and drivers.*
3. ***Flight service****: booking the tickets and determining the Going-back destination.*

4. ***Hotel Reservations:*** *determine the hotel, location of the hotel, number of rooms, number of days to reserve, and the day of arrival.*
5. ***Tour guide****: enter the country and city and select a tour guide.*
6. ***Travel companion:*** *enter age and language, when traveling alone, you may choose someone also alone to travel with you.*
7. ***Places of events and tourist attractions that need a reservation:*** *You specify the destination, so we show you the tourist attractions, then take you to the booking page and post the data.*
8. ***Customer Service****: It allows you to write what you need, questions, suggestions, or complaints, and they will help you.*

2.3 User Characteristics and Characteristics

Manager*: Administrator of the application.*
User*: end user wanting to benefit from our program.*
Airlines*: the airline companies we offer and work with.*
Employee*: The employees who work for us, such as the car driver, tour guide, and companion.*

2.4 Operating Environment

WayToGo *will be a software application accessible on any device with an internet connection, ensuring seamless access and convenience for users.*

2.5 Design and Implementation Constraints

The system should possess the following attributes:

- *A user-friendly interface, catering to staff with diverse technical proficiency.*
- *Implementation of robust security measures to safeguard patron data stored in the database.*
- *Dependency on an Internet connection for functionality.*
- *Requirement for travelers to be at least 18 years old and possess a valid passport.*
- *-It is difficult for the program to serve all countries.*
- *Continuous development, maintenance, and a substantial budget allocation for enhancing security systems.*

2.6 User Documentation

Comprehensive user manuals and online help guides will be readily available for both staff and travelers, ensuring they can navigate and utilize ***WayToGo*** *effectively and with confidence.*

2.7 Assumptions and Dependencies

WayToGo *will rely on a database to store crucial information such as travel tickets and hotel reservations. Additionally, it will be dependent on the Absher application for certain functionalities.*

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

*The application starts with a visually appealing welcome screen, see Figure 1 that introduces the user to the **WayToGo** platform and offers various options for authentication. Users can choose to continue with different authentication methods, such as email or a national single sign-on (NAFATH), or they can proceed as a guest. This initial screen sets a welcoming tone and helps users seamlessly enter the application.*
For new users, the sign-up process is facilitated through a user-friendly interface. The sign-up screen, see Figure 2 prompts users to provide their email or phone number, full name, and a password that meets specific requirements, such as containing at least six characters and including a number. This interface ensures a smooth onboarding experience and allows users to quickly create their accounts.
For returning users, the log-in screen, see Figure 3 provides a convenient way to access their existing accounts. Users can enter their registered email or phone number and password to log in. Additionally, options for remembering login details or recovering a forgotten password see Figure 4 may be available to enhance usability and account security.

re 1-Welcome page.

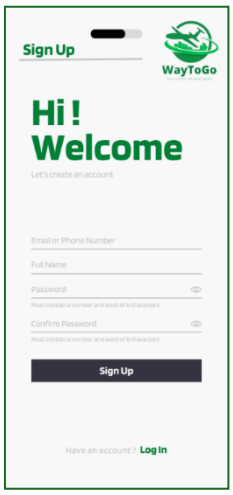


Figure 2-Sign up page.

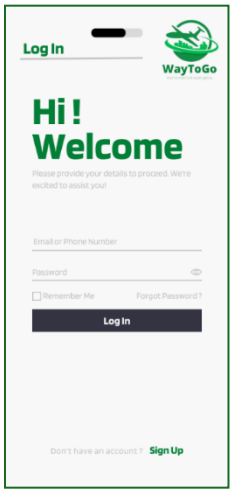


Figure 3-Log in page.

Figure 4-password recovery.

Travel Categories:
*The **WayToGo** application offers a range of travel-related categories, such as hotels, flights, car rentals, and attractions see Figures5-9. Each category is presented through a well-structured interface that enables users to search for their desired options easily. Users can input their preferences, including the destination, dates, and the number of travelers. Additional features, such as filtering options, special offers, and user profiles, may enhance the overall user experience within each category.*

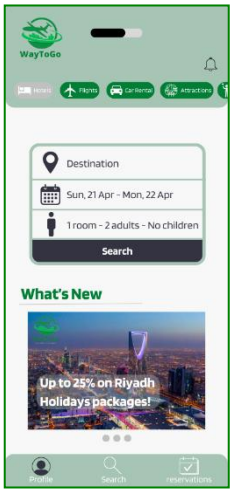
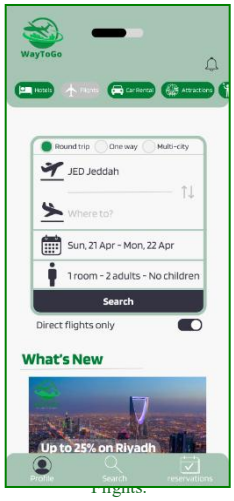
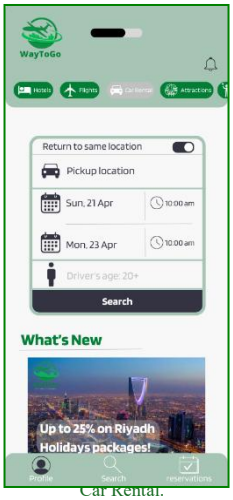


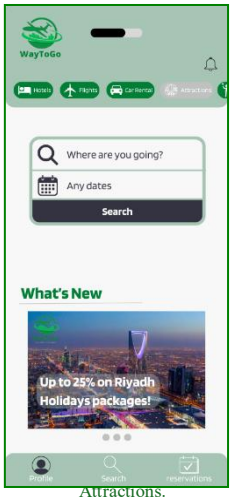
Figure 5-Search page/ Hotels.



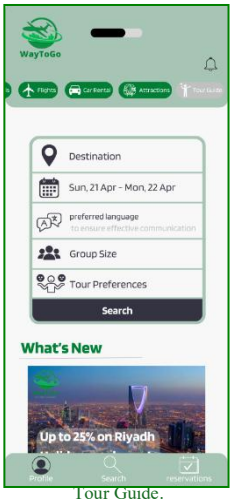
Flights.



Car Rental.



Attractions.



Tour Guide.

User Profile and Account Management:

The user profile screen see Figure 10 , Figure 11 provide users with a comprehensive interface to manage their personal information and account settings. Users can view and update their profile details, including their name, gender, and date of birth. Contact information, such as email address and phone number, can also be managed through this interface. Furthermore, users can customize their language preferences, set their preferred currency, control notification settings, and font size adjustment functionality to support visually impaired users. Furthermore, users can customize their language preferences, set their preferred currency, control notification settings, and. The ability to add payment methods and reset passwords for account security is also available within this section.

Reservation Tracking:

*To keep users informed about their travel plans, the **WayToGo** application includes a reservations screen see Figure 12. This screen displays users' current and past reservations in an organized manner, providing essential details such as dates, locations, and trip information. Users can easily access and manage their bookings from this screen, ensuring a seamless travel experience.*

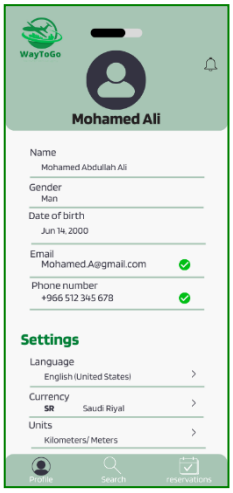


Figure 10-Profile page.

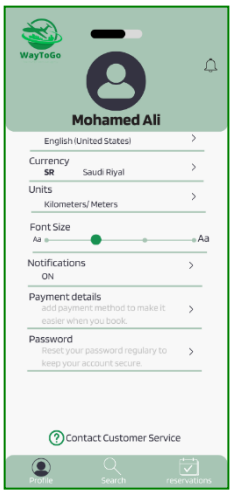


Figure 11-The rest of Profile page.

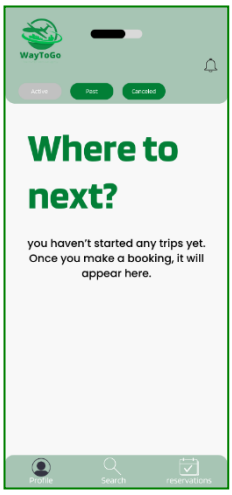


Figure 12-Booking History.

3.1.2 Hardware Interfaces

The application is exclusively compatible with iOS devices and requires multiple servers, high RAM, and storage space in terms of hardware requirements. It is also sensitive to facial recognition for access through (NAFATH).

3.1.3 Software Interfaces

The application supports iOS version 17.4.1 and seamlessly integrates GPS functionality, allowing for the determination of the user's current location. It also provides real-time updates on the flight's progress and accurate estimated time of arrival. The application code is written in Java SE version 22.0.1 .

3.1.4 Communications Interfaces

The application features several communication interfaces to facilitate seamless interaction between users and the booking system. Firstly, the Email Interface allows users to effortlessly stay updated on their bookings by receiving confirmations and flight updates directly to their email inbox. Additionally, users can conveniently communicate with customer support teams, as the interface enables them to respond to inquiries and make booking changes via email. Secondly, the Network Interfaces ensure smooth communication between the application and servers by supporting widely used protocols like HTTP/HTTPS. This ensures reliable data transmission and efficient operation of the booking system. Moreover, the Messaging Interfaces provide users with a direct channel to interact with customer support teams, allowing them to seek assistance and resolve any booking-related issues promptly. To safeguard user data and ensure privacy, the application employs Communication Security measures such as encryption using SSL/TLS protocols. Lastly, Data Transfer Rates are optimized to ensure high-speed data uploads, contributing to a seamless user experience while using the application. Together, these communication interfaces enhance user convenience, security, and overall satisfaction with the booking process.

3.2 Functional Requirements

3.2.1 < Sign-In / up >

3.2.1.1 Introduction

*- Enable users to access their accounts within the **WayToGo** travel application using different authentication methods.*

3.2.1.2 Inputs

- *Accept username and password, as **well as Absher authentication.***

3.2.1.3 Processing

- *Verification: The system verifies the entered username and password against the stored credentials in the database.*
- *Authentication: If the entered credentials are valid, the user is granted access to the **application and their personalized account.***
- *Authenticate through Absher if selected.*

3.2.1.4 Outputs

- *Successful Sign In: The user gains access to their account and the application's features and functionalities.*
- *Error Message: In case of invalid credentials or other authentication issues, an error message is displayed, indicating the reason for the failure to sign in.*

3.2.1.5 Error Handling

- *Handle scenarios such as incorrect username or password, failed Absher authentication, duplicate accounts during sign-up, password requirements, and password recovery.*
- *Handle scenarios such as incorrect usernames or passwords, failed Absher authentication, duplicate accounts during sign-up, password requirements, and password recovery.*

3.2.2 < Search >

3.2.2.1 Introduction

- The search function allows users to find relevant information related to their travel needs. It provides a page that contains various travel categories such as hotels, flights, current attractions at the destination, and tour guides for the trip. Instead of typing search queries, the user can select one of the available categories to retrieve specific information.

3.2.2.2 Inputs

- *Selected travel category (e.g., hotels, flights, attractions, tour guides).*

3.2.2.3 Processing

- *Retrieve the selected category from user input.*
- *Access the appropriate database or external API to fetch relevant information based on the selected category.*

- *Apply any necessary filters or sorting algorithms to refine the results.*

3.2.2.4 Outputs

- *Display a page to present relevant information related to the selected travel category.*
- *Show details such as hotel options, flight availability, attractions at the destination, or tour guide profiles.*

3.2.2.5 Error Handling

- *Handle cases where no search results are found for the selected category.*
- *Provide appropriate error messages or notifications to the user.*
- *Offer alternative options or suggestions if the search does not yield any results.*

3.2.3 < Booking Hotel >

3.2.3.1 Introduction

- *This function allows for booking accommodation based on the type of lodging (hotel room -apartment - villa), the number of individuals and the specified budget.*

3.2.3.2 Input

- *Number of individuals*
- *Budget*

3.2.3.3 Processing

- *The customer's selected lodging data is linked with the available times and categories from lodging companies. Afterward, what is available is displayed with lodging details (available duration of stay, accommodation price, type of lodging). If it aligns with the customer's preferences, the reservation is confirmed, and the invoice is presented with the available payment method to complete the payment process. The accommodation is then booked for the customer.*

3.2.3.4 Output

- *Displaying the reservation confirmation along with the details of the accommodation location*

3.2.3.5 Error Handling

- *Payment Issues: If there are issues with payment (e.g., card declined), the system should prompt the user to try a different payment method.*
- *Technical Errors: If there are technical problems (e.g., server down), the system should inform the user to try again later.*
- *Confirmation Problems: If there are issues confirming the booking, the system should guide the user on how to proceed or contact support for help.*

3.2.4 < Flight Service >

3.2.4.1 Introduction

- The flight service function manages the booking of tickets and determines the return destination for customers.

3.2.4.2 Inputs

- *Customer information (name, contact details).*
- *Departure location.*
- *Departure date and time.*
- *Return date and time (optional).*
- *Passenger count.*
- *Preferences (e.g., class, meal preferences).*
- Collect customer inputs.

3.2.4.3 Processing

- *Collect customer input.*
- *Check availability of flights based on departure and return dates.*
- *Offer available flights to the customer.*
- *Confirm booking based on customer preferences.*
- *Determine the return destination based on customer's departure location.*

3.2.4.4 Outputs

- *Confirmed booking details (flight numbers, departure/arrival times, seat assignments).*
- *Return destination for round-trip bookings.*

3.2.4.5 Error Handling

- *If the departure date provided is invalid or unavailable, the system notifies the customer and requests a valid date.*
- *If a return date is provided but is invalid or unavailable, the system notifies the customer and suggests alternative options.*
- *In cases of insufficient seat availability or other booking constraints, the system informs the customer and provides alternative solutions.*
- *Input errors are managed gracefully by the system, offering clear instructions and options for correction.*

3.2.5 < Renting a car >

3.2.5.1 Introduction

- The renting a car feature enables users to rent a car by providing the necessary information such as the location to receive the car, the location of its return, the duration of the rental, and the driver's data for legal compliance.

3.2.5.2 Input

- *Location to receive the car.*
- *Location of return*
- *Duration of rental*
- *Driver's data (name, contact information, driving license details, and any additional information required by the law or the car rental company)*

3.2.5.3 Processing

- *Validate the location to receive the car within the car rental service's coverage area.*

- *Validate the location of return within the car rental service's coverage area.*
- *Validate the specified duration of the rental and check for any restrictions or conflicts with car availability.*
- *Verify the driver's data to confirm eligibility to drive, including checking driving license validity, age restrictions, and any additional requirements set by the law or the car rental company.*

3.2.5.4 Output

- *Confirmation of the successful rental process, indicating that the car has been reserved for the specified location, duration, and driver's data.*
- *Rental agreement or booking details, which may include the rental period, car details, pick-up and return locations.*

3.2.5.5 Error Handling

- *If the provided location to receive the car or the location of return is invalid or outside the car rental service's coverage area, appropriate error messages should be displayed, and the user should be prompted to provide valid locations.*
- *If the specified duration of the rental conflicts with the availability of the car or violates any restrictions, the system should notify the user and suggest alternative options if available.*
- *If the driver's data provided by the user does not meet the legal requirements or any additional criteria set by the car rental company, the system should inform the user about the issue and request the necessary corrections or further information.*

3.2.6 < Booking Attractions >

3.2.6.1 Introduction

- In this service, customers have the opportunity to book an event for their special occasions or gatherings. The event booking service provides a convenient way for customers to secure a venue and necessary arrangements to ensure a successful event.

3.2.6.2 Input

- *Event Type: The type or nature of the event e.g. (birthday party, conference).*
- *Date and Time: The preferred date and time for the event.*
- *Venue: The desired location for hosting the event.*
- *Number of Guests: The estimated number of guests expected to attend the event.*
- *Budget: The customer's budget for the event.*

3.2.6.3 Processing

- *Venue Availability: The system checks the availability of the chosen venue for the specified date and time, Capacity: The system ensures that the selected venue can accommodate the estimated number of guests.*
- *Services and Amenities: The system provides information about the services and amenities offered by the venue.*
- *Cost Estimation: The system calculates the total cost of the event based on the chosen venue, services.*
- *Confirmation: the system confirms the booking and provides the customer with a confirmation receipt containing all the event specifics.*

3.2.6.4 Output

- *Booking Confirmation: The system generates a confirmation document that includes the event details, venue information, selected services, and total cost.*
- *Payment: The system presents available payment methods and instructions for completing the payment for the event booking.*
- *Payment Confirmation: After the payment is successfully processed, the system issues a payment confirmation.*

3.2.6.5 Error Handling

- *Venue Availability: If the chosen venue is not available, suggest alternative dates or venues.*
- *Capacity Limit: If too many guests are estimated for the venue, recommend a larger venue or reducing guest numbers.*
- *Payment Problems: Guide users through payment issues or offer alternative payment methods.*
- *Confirmation Errors: Inform users of any errors during confirmation and provide assistance.*
- *Technical Issues: Advise users to try again later if there are technical problems.*
- *Payment Issues: If there are issues with payment (e.g., card declined), the system should prompt the user to try a different payment method.*
- *Technical Errors: If there are technical problems (e.g., server down), the system should inform the user to try again later.*
- *Confirmation Problems: If there are issues confirming the booking, the system should guide the user on how to proceed or contact support for help.*

3.2.7 < Book a tour guide >

3.2.7.1 Introduction

- *"Book a tour guide" is a service that allows travelers to hire a local guide to show them around a specific destination, and this platform connects tourists with knowledgeable and experienced guides who can provide valuable insights and information about the places they visit.*

3.2.7.2 Processing

- *It will be possible to choose a male or female tour guide. After that, he will choose the guide's name and the time and day he wants. When the customer chooses the guide, it will be done automatically by going to the program's database and checking whether the guide is available or not. With the service of choosing a tour guide, the customer can also choose the payment method. Will it be active Credit card, Visa, or cash. After that, the user will be asked. He will be asked to approve his choice of guide, time, day, and method of payment. If he agrees to carry out the process, the money will be withdrawn from the customer and completed automatically, and the reservation will be successful.*

3.2.7.3 Input

- *Select name of guide*
- *Select time and day.*
- *Select payment method.*

3.2.7.4 Output

- *Tour guide information (name, gender, time, number of days).*
  Payment method (Service price, use credit card or visa or cash)

3.2.7.5 Error Handling

- *Guide Availability: If the selected guide is not available at the chosen time and day, the system should inform the user and prompt them to select a different guide or choose an alternative time/day.*
- *Payment Authorization: In case of payment method failure (e.g., insufficient funds for credit card payment), the system should notify the user and prompt them to try a different payment method or provide alternative payment details.*
- *Invalid Input: If the user provides invalid input (e.g., entering an invalid time or day format), the system should provide clear instructions on the required input format and prompt the user to correct their input.*
- *Transaction Errors: If there are any errors during the payment transaction (e.g., network issues), the system should inform the user and advise them to try again later or contact customer support for assistance.*
- *Confirmation Errors: If there are errors during the confirmation process (e.g., failure to generate a booking confirmation), the system should inform the user and provide instructions on how to proceed or contact support for further assistance.*

3.2.8 < Setting >

3.2.8.1 Introduction

- The setting function allows users to customize and manage their application preferences and configurations. It provides a page where users can modify various settings such as language, currency, units, font size, notification preferences, payment details, and passwords.

3.2.8.2 Inputs

- *User-selected settings and preferences for language, currency, units, font size, etc.*
- *Changes made to notification preferences, payment details, or passwords.*

3.2.8.3 Processing

- *Retrieve user input and selected settings.*
- *Process the changes made in the settings and configurations.*
- *Update the application's configuration or user profile with the modified settings.*

3.2.8.4 Outputs

- *Reflect the updated settings within the application.*
- *Apply the modified configurations to personalize the user experience.*
- *Display relevant notifications or feedback regarding the setting changes.*

3.2.8.5 Error Handling

- Handle cases where invalid settings or configurations are provided
- Validate user input to ensure it adheres to the specified format or restrictions
- Provide appropriate error messages or notifications if the settings cannot be applied
- Offer guidance or suggestions to help the user correct any invalid settings or resolve any errors

3.2.9<Profile>

3.2.9.1 Introduction

- *The "Profile" function allows users to manage and access their personal profiles within the **WayToGo** travel application. It provides a dedicated space for users to view and update their personal information, including their name, gender, date of birth, email, and phone number.*

3.2.9.2 Inputs

- *Profile Update: Users can input or modify their personal information, including the following fields (Name, Gender, Date of Birth, Email, Phone Number).*

3.2.9.3 Processing

- *Profile Retrieval: Once authenticated, the system retrieves the user's profile information from the database.*
- *Profile Update: If the user chooses to update their profile, the system processes and saves the modified information to the database.*

3.2.9.4 Outputs

- Profile Information: The system displays the user's profile information, including their name, gender, date of birth, email, and phone number.
- Profile Update Confirmation: After successfully updating the profile, the system notifies the user about the successful modification and confirms the changes made.

3.2.9.5 Error Handling

- *Profile Not Found: In case the user's profile information is not available or cannot be retrieved from the database, the system displays an error message informing the user about the issue and suggests contacting customer.*
- *support Invalid Profile Update: If the user inputs invalid or incomplete data while updating their profile, the system displays an error message and prompts the user to provide valid information.*

3.2.10 < Reservation Tracking >

3.2.10.1 Introduction

- *The Reservation Tracking feature in the **WayToGo** application informs users about their travel plans by providing a reservation screen. This screen displays users' current and past reservations in an organized manner, including essential details such as dates, locations, and trip information. Users can conveniently access and manage their bookings from this screen, ensuring a seamless travel experience.*

3.2.10.2 Input

- *Reservation data: The system should have access to the user's reservation data, including reservation details such as dates, locations, and trip information.*

3.2.10.3 Processing

- *Retrieve reservation data: The system retrieves the reservation data from the database or designated storage location.*
- *Organize reservations: The system organizes the reservations in a user-friendly manner, grouping them by current and past reservations and sorting them by date or relevant criteria.*

- *Display reservation details: The system displays essential details of each reservation, such as dates, locations, and trip information, on the reservations screen.*
- *Enable reservation management: The system provides functionality for users to easily access and manage their bookings, including viewing reservation details, modifying reservations (if allowed), and canceling reservations (if permitted).*

3.2.10.4 Output

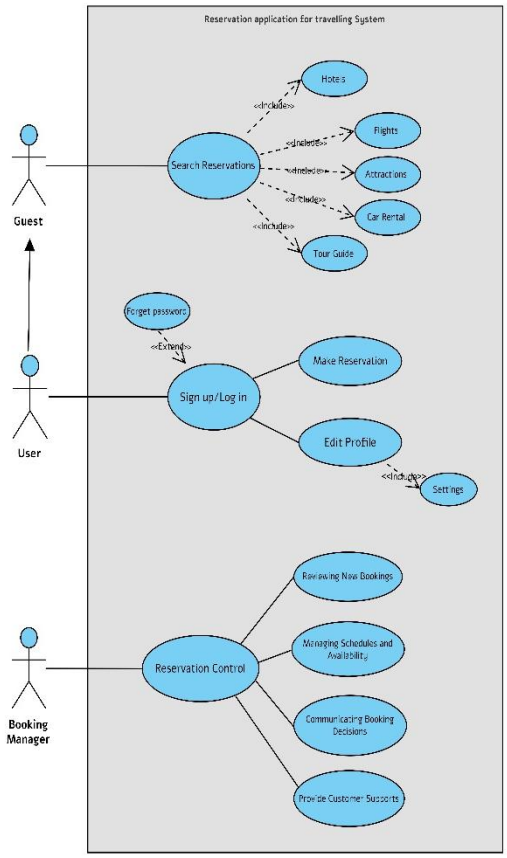
- *Reservations screen: The system presents a screen that displays users' current and past reservations in an organized manner.*
- *Reservation details: The system displays essential details, such as dates, locations, and trip information, for each reservation on the screen.*
- *Reservation management options: The system provides options for users to interact with their reservations, including viewing details, modifying bookings, or canceling reservations if applicable.*

3.2.10.5 Error handling:

- *If there are no reservations associated with the user, the system displays a message such as "Where to next? ". Once you make a booking, it will appear here.*
- *If there are no reservations associated with the user, the system displays a message such as "Where to next?. Once you make a booking, it will appear here."*
- *If there are any errors in retrieving or processing the reservation data, the system displays an error message and provides options for the user to retry or seek assistance.*

3.3Use Cases

3.3.1 Use Case #1



3.4 Non-Functional Requirements

3.4.1 Performance

- *The application should respond quickly to user interactions, providing a seamless and responsive experience.*
- *It should handle a large number of concurrent users without significant performance degradation.*

3.4.2 Reliability

- *The application should be stable, minimizing system crashes or unexpected errors.*
- *It should have appropriate backup and recovery mechanisms to ensure data integrity.*

3.4.3 Availability

- *Users can easily adjust font sizes within the app for better readability.*
- *Unit measurements (e.g., kilometers, miles) can be changed based on user preferences.*
- *Currency settings automatically adjust based on the user's location or can be manually customized.*
- *The app stays usable during backup and recovery, avoiding disruptions for users.*

3.4.4 Security

- *It should have appropriate authentication mechanisms and enforce password strength requirements.*
- *The application should implement robust security measures to protect user data, including encryption of sensitive information and secure transmission protocols.*

3.4.5 Maintainability

- *Schedule routine maintenance, updates, and patches during off-peak hours to minimize user disruption during busy periods.*
- *Design the password recovery process to be modular and easy to maintain for future enhancements.*

3.4.6 Portability

- *Utilize intuitive icons and clear titles to enhance navigation and usability, making it effortless for users to identify and access different sections within the app.*
- *Design the app to adapt smoothly to various screen sizes and orientations, providing a user-friendly experience on smartphones and tablets.*

3.4.7 Integrity

- *Implement strong security measures to protect user data and ensure the accuracy and reliability of information stored in the app.*
- *Prevent unauthorized access and maintain data integrity throughout the password reset process.*

3.5 Inverse Requirements

- *The application shall not display personal identification information, such as passport numbers, to unauthorized users.*
- *The system shall not store credit card CVV codes in any form after processing transactions.*
- *The application shall not allow bookings without valid payment authorization.*
- *The system shall not provide access to reservation details to users who are not associated with the booking.*
- *The application shall not send promotional messages to users who have unsubscribed from marketing communications.*

3.6 Design Constraints

3.6.1 Usability Constraints:

- The system must provide clear and concise error messages to users.
- The system must offer context-sensitive help to guide users through tasks.
- User input fields must be clearly labeled and easily accessible.